7.6.2 Symptoms

Figure 7.13 Cherry red spot in the retina

Muscle weakness, progressive retardation in development and difficulty in eating are typical early symptoms. Mental retardation and blindness are the characteristic symptoms in this rare genetic disorder. Death between 3 -4 years is unavoidable. More than 90 % of the patients have a characteristic cherry red spot in the retina. Tay-Sachs disease can be diagnosed by taking amniotic fluid from the motherand assaying the hexosaminidase A activity. Bad memory, defective learning skills, poor decision making, eventually the child goes blind, deaf, paralyzed, and dies within 3 years.

Till now, there is no cure for Tay-Sachs disease. Treatments include gene therapy or enzyme replacement therapy. They are not promising cure for the disease but could slow down disease progression.

SUMMARY

Metabolic processes, including transport of molecules across the membrane, are coordinated by a series of biochemical reactions and each reaction is controlled by an enzyme. One or both of a pair of genes involved in the production of a particular enzyme may be defective. The production of an enzyme, be it normal or low, depends on whether the gene is dominant or recessive, which is inherited from the parents. Inborn errors of metabolism, are fortunately, linked to recessive traits. Though, the consequences of this disease are extremely rare, they are collectively common and presented in new born infants or shortly thereafter.

The basis of IEM is the mutation in a gene that codes for a particular enzyme in a specific pathway, involving either the break-down or storage of carbohydrates, fats and proteins.

1. The term 'inborn errors of metabolism' was first proposed by Garrod followed by Beadle and Tatum
2. Common defects in biochemical pathways include, defects in the transport, excessive accumulation of substrate, deficiency of product and secondary inhibition.
3. Enzymes play a major role in performing the anabolism and catabolism of reactive pathways and serve as catalysts in the conversion of one compound to the other.
4. The major classification of inborn errors of metabolism includes protein, carbohydrate,

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